MA'ADEN معادن

MA'ADEN ENGINEERING MANUAL

Wastewater Treatment Plant

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1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for design, supply and installation of a Package Wastewater Treatment Plant (WWTP) to provide sanitary wastewater treatment for all Ma'aden projects.

The specification:

- Identifies applicable Standards and guidelines to be used for the Work
- Sets the minimum performance requirements of a package WWTP
- Defines the general requirements for the Work.

This document is intended to be used in conjunction with the referenced documents listed herein.

1.2 Purpose

This document is to be used as a reference for the preparation of a particular package Wastewater Treatment Plant specification. Such specifications shall meet the requirements specified within this document as a minimum.

2. KEY TERMS AND DEFINITIONS

ABS	Acrylonitrile Butadiene Styrene
ADWF	Average Dry Weather Flow (usually expressed in m^3/d)
BOD	Biochemical Oxygen Demand (mg/L)
Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform Work or cause Work to be performed.
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
FAT	Factory Acceptance Test
HSEC	Health, Safety, Environment and Community
ITP	Inspection and Test Plan
JSA	Job Safety Analysis
MCC	Motor Control Center
MSDS	Material Safety Data Sheet
OIP	Operator Interface Panel
PCS	Process Control System

PDWF	Peak Dry Weather Flow
PFD	Process Flow Diagram
P & ID	Piping and Instrumentation Diagram
PLC	Programmable Logic Controller
PME	Presidency of Meteorology and Environment
PPE	Personal Protective Equipment
RCER	Royal Commission Environmental Regulations
SAT	Site Acceptance Test
SOP	Standard Operation Procedure
SS	Suspended Solids (mg/L)
TN	Total Nitrogen (mg/L)
TP	Total Phosphorus (mg/L)
TSS	Total Suspended Solids (mg/L)
UPS	Uninterruptible Power Supply
Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.
WWTP	Wastewater Treatment Plant

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All Work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All equipment and systems shall conform to the following:

3.2.1 Local Codes and Standards

- Royal Commission Environmental Regulations
- Royal Commission Environmental Regulations – Penalty System
- Saudi Building Code SBC701 – Sanitary
- Saudi Building Code SBC401 – Electrical.

The Contractor shall also comply with the following directives of the Ministry of Interior, High Commission for Industrial Security – Kingdom of Saudi Arabia:

- Safety and Fire Protection Directives for Industrial Facilities
- Security Directives for Industrial Facilities.

3.2.2 International Codes and Standards

3.2.2.1 EPA – The U.S Environmental Protection Agency

3.2.2.2 BS – British Standard

BS 4994	Specification for design and construction of vessels and tanks in reinforced plastics

3.2.2.3 ANSI – American National Standard Institute

ANSI ASC A14.3	American National Standards for Ladders – Fixed – Safety Requirements
ANSI Z400.1	Hazardous Workplace Chemicals – Hazard Evaluation and Safety Data Sheet and Precautionary Labeling Preparation

3.2.2.4 ASSE – American Society of Safety Engineers

ASSE A1264.1	Safety Requirements for Workplace Walking/Working Surfaces and Their Access; Workplace, Floor, Wall and Roof Openings, Stairs and Railing Systems

ASSE Z117.1	Safety Requirements for Confined Spaces
ASSE Z359.1	Safety Requirements for Personal Fall Arrest Systems, Subsystems and Components

3.2.2.5 International Code Council (ICC)

ICC EC	ICC Electrical Code – Administrative Provisions
ICC IBC	International Building Code
ICC IEBC	International Existing Building Code
ICC IECC	International Energy Conservation Code
ICC IFC	International Fire Code
ICC IMC	International Mechanical Code
ICC IPC	International Plumbing Code

3.2.2.6 American Society of Mechanical Engineers, ASME International (ASME)

ASME B16.5	Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24 – Metric/Inch Standard
ASME B31.1	Power Piping

3.2.3 Alternative Codes

Alternative codes may be used if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-AR-SPC-0007	Project Site Offices (Temporary)
MD-101-SMEM-EG-CS-SPC-0004	Plant Control System (PCS) – Network Equipment
MD-101-SMEM-EG-CV-SPC-0001	Earthworks
MD-101-SMEM-EG-CV-SPC-0002	Anchor Bolts
MD-101-SMEM-EG-CV-SPC-0007	Geomembranes
MD-101-SMEM-EG-EL-SPC-0001	Electrical and Instrumentation Installation
MD-101-SMEM-EG-EL-SPC-0002	Electrical & Instrumentation Calibration, Inspection & Testing
MD-101-SMEM-EG-EL-SPC-0004	Electric Motors
MD-101-SMEM-EG-EL-SPC-0007	Variable Speed Drives
MD-101-SMEM-EG-EL-SPC-0008	Motor Control Centers (MCCs)
MD-101-SMEM-EG-FI-SPC-0002	Fire and Gas System – General
MD-101-SMEM-EG-IN-SPC-001	Instrumentation

MD-101-SMEM-EG-ME-SPC-0005	Materials Handling
MD-101-SMEM-EG-ST-SPC-0001	Steel Fabrication
MD-101-SMEM-EG-ST-SPC-0002	Steel Erection
MD-101-SMEM-EG-ST-SPC-0005	Concrete Works
MD-101-SMEM-EG-ST-SPC-0008	Protective Coating for Concrete
MD-101-SMEM-EG-ST-SPC-0009	Protective Coating of Structural Steelwork
MD-101-SMEM-EG-SY-SPC-0002	Emergency Warning System
MD-101-SMEM-EG-TE-SPC-0003	Telephone System
MD-101-SMEM-EG-TE-SPC-0004	Radio Equipment
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

3.4 Site Data

All materials, equipment and work methods shall be suitable for the Site conditions as defined in the applicable Site Data Specification.

4. UNITS OF MEASUREMENT

The units of measurement used in this specification are primarily those recommended by the International System of Units (SI), but also include:

- Non-SI units that are based on the SI and are retained because of their practical importance and common usage (e.g. hectare, liter and tonne)
- Non-SI units that are retained because of their practical importance (e.g. day, hour, minute and degree Celsius).

The units of measurement in this specification are as follows:

a	year (annum)	c (prefix)	centi- (SI prefix = 0.01)
d	Day	g	gram
g/L	grams per litre	g/mL	grams per milliliter
h	Hour	kg	kilogram
kW	Kilowatt	L	liter
L/p.d	liters per person per day	L/s	liters per second
m	Meter	m^2	square meter
m^3	cubic meter	m^3/d	cubic meters per day
m^3/h	cubic meters per hour	mg	milligram
mg/L	milligrams per liter	mL	milliliter
mm	Millimeter	MPa	megapascal
NTU	Nephelometric Turbidity Units	s	second
°C	degree Celsius		

5. MATERIALS AND EQUIPMENT

5.1 Equipment Selection and Design

Equipment selected and/or designed for the WWTP shall be 'fit-for-purpose' and shall meet the requirements of this Specification as a minimum. All such equipment shall be clearly shown on WWTP design drawings and equipment schedules submitted to the Employer for approval.

5.2 Selection against Corrosion

The Contractor shall select materials and equipment that are capable of withstanding corrosion in the exposure condition for which they are intended.

All materials, chemical feed equipment and storage facilities shall either be constructed of corrosion-resistant materials or have a protective coating. Construction shall meet the general requirements for corrosion free materials.

Materials shall be selected that are resistant to corrosion and any other forms of deterioration. Usually in case of corrosion, consideration is given to the possible presence of gases such as hydrogen sulfide, other corrosive gases, greases, oils and similar constituents that are usually found in sanitary wastewater.

Any material or equipment showing signs of corrosion during the maintenance period shall be rejected and replaced by the Contractor at his own expense with material or equipment resistant to corrosion as shown by a re-test.

5.3 Material Deliveries

The Contractor shall be responsible for off-loading all the materials and equipment at the Site as well as the storing of such materials in a depot as nominated by the Employer.

The Contractor shall be solely responsible for obtaining transport permits and clearance (if required) from road or rail authorities for the transporting and delivery of materials and equipment as well as the compliance with the requirements of such bodies.

The Contractor shall be responsible for the supply, delivery, installation and commissioning of the following:

- The treatment unit together with all the mechanical equipment, valves, pumps, interconnecting pipework, structural work, foundations work and electrical work necessary for the correct functioning of the whole
- All electrical equipment on the distribution side of a power supply point, and
- Any other equipment included in the quotation by the Contractor.

5.4 Piping, Valves and Fittings

Piping, valves and fittings selections for the proposed system shall be in accordance with relevant codes and standards referenced in Section 3.2 and Section 3.3. The Contractor shall nominate the valves, piping materials and joining methods for each service for approval by the Employer.

6. FUNCTIONAL REQUIREMENTS – SANITARY WASTEWATER TREATMENT

6.1 General Requirements

6.1.1 Design and Certification

The following general requirements shall apply:

- All structures, equipment and auxiliaries shall be designed for a service life as specified in Table A1 in Appendix A. Where the intended design life cannot be reasonably met, or is not practical for certain components, the design shall allow for these items to be readily replaced. The design shall be subject to final review by the Employer before the Contractor proceeds with fabrication.
- All equipment and auxiliaries shall be suitable for 24-hours outdoor operation unprotected from the environment. If enclosures are necessary to protect equipment, these shall be provided by the Supplier.
- All components of the WWTP shall be new and constructed from new materials
- The WWTP shall be capable of operating continuously 24 hours per day, 7 days per week and shall be automatic in operation with provision for manual control of all operations and adjustment of timers and other process operations
- All wastewater treatment equipment supplied shall:
  - Operate reliably during the design life with minimal down time for maintenance
  - Perform reliably under the anticipated range of environmental conditions, flow quantity and quality, and operating conditions
  - Operate without generation of nuisance odour and noise, and
  - Contain generic parts that can be replaced easily "off the shelf" when required.
- All components shall be manufactured from sufficiently durable materials to allow for operation in direct sunlight and at a range of temperatures, typically from 15°C to 45°C
- All fittings and pipework shall be ABS or stainless steel unless approved otherwise by the Employer
- The Contractor shall provide relevant calculations and formal sign-off by a Chartered Structural Engineer as certification that the WWTP and all components thereof satisfy all structural requirements.

6.1.2 Proposed Treatment Processes

To ensure that the treated wastewater meets the required discharge standards the system configuration shall be set up as follows:

- The WWTP shall cater for a peak population as specified in Table A-1 in Appendix A
- The WWTP shall be designed to meet all the functional requirements stated within this document with emphasis placed on meeting the treated wastewater quality standards as specified in Section 6.1.7
- The WWTP plant shall be located in the proximity of the services area at each Site, as shown on the drawings.

6.1.3 Inlet Screening Facilities

The inlet works shall include a bypass screening unit. This unit shall be designed to provide sufficient protection for downstream equipment. The design shall ensure prevention of short-circuiting of raw wastewater to the "clean"/screened side during normal operating conditions.

6.1.4 Plant Inflow Streams and Design Loading

The WWTP shall be designed as per the design loading given in Table A-1 in Appendix A.

6.1.5 Plant Configuration, Operation and Mode of Disposal

The proposed plant configuration and mode of disposal of treated wastewater from the WWTP shall be as detailed in Table A-2 in Appendix A.

Operation of the WWTP shall be controlled automatically. The software shall be designed to optimise the plant in any one of the operating modes, and maximise operating flexibility in order to provide the best performance.

6.1.6 Sludge Disposal

Design of the WWTP shall incorporate all necessary sludge disposal, treatment and/or storage facilities required to ensure that de-sludging is not required at a frequency greater than once per month unless advised otherwise by the Employer. The WWTP shall therefore provide a minimum sludge storage capability of one month or as advised by the Employer without any adverse effect on the guaranteed treated wastewater quality.

Details of the sludge volume and characteristics shall be provided in addition to proposed disposal, treatment and/or storage facilities. This information shall be provided within the Contractor's proposed works and equipment specifications.

6.1.7 Required Treated Wastewater Quality

Treated wastewater shall comply with the discharge standards specified in Royal Commission Environmental Regulations, Kingdom of Saudi Arabia and any other particular standards identified and specified for a project.

6.1.8 Control and Instrumentation

All stages of treatment shall be controlled by a single PLC. The PLC shall be designed to operate in "stand alone" mode and be capable of operating without dependency on communications with other plant and/or systems. A single OIP shall be locally installed on the WWTP switchboard to enable drive and equipment monitoring.

All systems and instruments shall be Yokogawa Smart Instruments (HART or Foundation Fieldbus). Where Yokogawa instruments are not available the instruments shall be sourced through Yokogawa to ensure consistency of logistics.

In the event that the Contractor is unable to meet with the requirements detailed above, the Contractor shall obtain written approval from the Employer with regards to any exceptions to this clause.

All control and power cables shall be taken to on-skid marshalling boxes before connection to the PLC and MCC. This is to allow for the future cut over to the Yokogawa PCS.

A UPS shall be installed to provide a continuous two (2) hour backup power supply to the PLC control platform. The UPS shall have automatic changeover capability.

Ma'aden Specifications for Plant Control System and Instrumentation as referenced in Section 3.3 shall be complied with.

6.2 Unit Process Requirements

6.2.1 Wastewater Treatment Package Plant

The WWTP should be in (an) enclosed structure(s) and placed on a suitable base consisting of a concrete slab.

The WWTP system shall consist of all tanks, pipes, valves, pumps, controls and other auxiliary equipment necessary for the effective treatment of the wastewater stream as per Section 6.1.7.

Any emergency overflows required within the WWTP shall overflow directly to the lined evaporation pond (detention pond) for storage prior to eventual return to the WWTP and further treatment.

The system operation shall be as flexible as possible and engineered for minimal operator attendance. The process equipment shall be easily accessible.

6.2.2 Odor Control

The WWTP shall be designed with the objective of preventing odor nuisance in the areas surrounding the WWTP. Odor impacts should be estimated and appropriate mitigation measures incorporated into the WWTP design to comply with the Ambient Air Quality Standards and General Air Quality Regulations specified in RCER, Kingdom of Saudi Arabia.

Odor control shall be achieved actively by covering process units and withdrawing the malodorous air.

Consideration shall be given to either removing the odor using a biofilter/carbon filter or alternate treatment before discharging to the atmosphere or simply discharging directly to the atmosphere in such a manner as to achieve effective odor dispersion.

6.2.3 Emission of Contaminants

The WWTP design shall comply with Section 2 – Air Environment of RCER, Kingdom of Saudi Arabia.

6.2.4 Noise Control

The WWTP design shall comply with the Environmental Noise Standards specified in the RECR, Kingdom of Saudi Arabia. The Contractor shall ensure that the maximum noise limit is met by the provision of noise attenuation device(s), which shall not interfere with the daily operation and maintenance of the WWTP.

The WWTP and other ancillary equipment shall comply with the relevant Environmental Protection (Noise) Regulations issued by the Presidency of Meteorology and Environment (PME).

7. INSTALLATION REQUIREMENTS

7.1 Signage and Labeling

The Contractor shall supply and install all mandatory caution, prohibition, safety and danger signage as required by the legislation. Signage shall be installed at all appropriate locations around the treatment plant.

All mechanical process equipment associated with the facility shall be labeled.

All electrical equipment shall be marked with an equipment tag. Equipment numbers will be provided following submission of the motor data sheets.

Above-ground pipework, conduits and ducts shall be provided with identification markers indicating contents and direction of flow in accordance with the relevant Ma'aden Specification referenced in Section 3.3. Markers shall be approved self-adhesive labels.

7.2 Hazardous Materials

All chemicals/reagents and their corresponding Material Safety Data Sheets (MSDS) shall be supplied with the WWTP and shall be pre-approved by the Employer prior to transportation to the Site.

The Contractor shall provide copies of MSDS to the Employer prior to delivery of any chemicals/reagents to the Site.

7.3 Spares

Appropriate spare equipment shall be recommended based on a 12 months operation of the WWTP.

8. OPERATION AND MAINTENANCE

The WWTP operation shall be automated to the maximum practical extent, to minimize the need for operator intervention. The Plant shall operate automatically, safely and efficiently during normal conditions, and recover or shutdown as appropriate if an undesirable or dangerous condition occur. The Plant shall be capable of operating for long periods with no operator intervention other than routine monitoring. The Contractor shall provide the details of operator training requirements, expected interventions, and man-hours required to operate and maintain the WWTP.

If the Contractor needs to monitor key process parameters from an offsite location, he shall liaise with the Employer to determine the relevant protocols and implementation requirements. Offsite monitoring systems shall not be implemented without the Employer's approval.

9. ONGOING PERFORMANCE

To certify that the design meets ongoing treated wastewater quality requirements, testing shall be undertaken by Others onsite for the parameters listed in Table 9-1. Failure of the WWTP to meet the quality criteria, operating under the designed loadings and within the warranty period, will result in the WWTP being deemed defective.

Table 9-1 : Treated Wastewater Monitoring Program

Quality Characteristics	Monitoring Frequency
5-day BOD (uninhibited)	Monthly
pH	Monthly
Disinfection Residual	Monthly
Faecal Coliforms, based on the average of a minimum of five samples collected	Monthly

10. INSPECTION AND TESTING

Operational and functional tests shall be conducted prior to dispatch of the WWTP to the Site.

An Inspection and Test Plan (ITP) shall be submitted by the Contractor for Employer's review and approval prior to the commencement of material and equipment fabrication and/or assembly. The submitted ITP shall include requirements for sub supplier equipment and shall identify all activities up to and including final Factory Acceptance Tests (FAT).

Contractor's ITP should aim to maximize testing activities completed off-site prior to installation.

All modular components that connect directly (without interconnecting piping) shall be trial fit in the Supplier's shop.

The Contractor shall submit Site Acceptance Test (SAT) procedures for Employer's review and approval prior to handover. These SAT procedures will be used by the installation Contractor to verify equipment is installed and tested to the satisfaction of the Employer.

11. SUBMITTALS

The following documentation shall be submitted to the Employer at appropriate stages of the design, supply, installation, testing and commissioning of the WWTP.

- Design drawings
- Calculations, PFD and P&IDs
- A detailed operation and maintenance manual for the supplied WWTP containing engineering data including the following:
  - Power supply requirements
  - Voltages and amperage ratings
  - Motor list
  - Nominated treated wastewater quality.
- General Arrangement drawings

- Installation procedure drawings
- "As Constructed" drawings
- Structural certification – refer to Section 6.1.1
- Training package
- Commissioning procedures
- Certification that the WWTP is fully operational and will meet the target discharge standards.

Appendix A

Specification – Required Data

Table A-1 : Design Loading

Required Service Life	To be determined for each project	years
Peak Population	To be determined for each project	persons
Avg. Per capita sewage flow	To be determined for each project	l/p.d
Diurnal/Shift Change Peaking Factor	To be determined for each project	-
BOD_5 loading	To be determined for each project	g/p.d
Total Nitrogen (TN)	To be determined for each project	g/p.d
Total Phosphorus (TP)	To be determined for each project	g/p.d
Suspended Solids (SS)	To be determined for each project	g/p.d
Wastewater Temperature	To be determined for each project	°C

TableA-2: Plant Configuration –WWTP

Plant Configuration	**Mode of Disposal**
Wastewater Treatment Plant (WWTP) – [1]ADWF (m^3/d); [2]PDWF (m^3/h)	To be determined for each project
Mode of disposal of treated wastewater	To be determined for each project

[1] To be replaced with a numeric value determined for the project
[2] To be replaced with a numeric value determined for the project